The primary aim of this project was to create a rule-based expert system that could guide the user towards an appropriate selection of laptop considering their personal preferences.

The system is programmed with the technique of forward chaining, which means it starts with the user-given set of facts and then applies the rules to derive new conclusions until there are no more rules to fire.

The program is designed to gather details about the users' budget for a laptop, how mobile they need the laptop they select to be, battery life needs, gaming needs, creative work requirements, and their OS preference, asking users a series of questions.

The obtained replies are then transformed into facts used by an inference engine to judge rules, which are expressed here in terms of IF, THEN conditions.

Our rule-based system works through the process of forward chaining, whereby it starts from some given facts, then the rules are applied in a top-down fashion, leading towards new conclusions which may trigger other rules if conditions for them are also satisfied. This goes on until there are no more rules left to fire.

There are 10 rules, presented in the form of "IF... THEN...", written by experts, which are stored in a file called laptop_rules.json.

Rules contain conditions, a decision, and a priority value. Priority determines how soon that rule is processed, while the conditions are the set of user facts that must hold for that particular rule to be processed.

The engine is implemented as a Python module engine.py, and it consists of three functions:

- can_fire() this function is used to check whether the current facts fulfill a given rule's conditions.
- run()- this method runs through applicable rules over and over again, firing them and adding new facts
- conclusions() it is this function that takes the conclusions and turns them out in the form of recommendations, specifications, and other facts.

Besides that, the inference engine also maintains which rules were fired, so as to explain the recommendations later on.

Test cases for six different scenarios were implemented by employing the unit-testing framework of Python.

The scenarios include premium ultrabooks, budget ultrabooks, gaming laptops, high-end gaming laptops, creative work requirements, and Linux-compatible hardware, and the tests cover these topics.

Our system was tested via those 6 scenarios, and it passed all the tests, showing that the conclusions made by the system were the ones that we predicted for the various scenarios.

In light of these tests, this project has shown that expert systems can be a great way to provide recommendations based on a set of facts, rules, and a reasoning engine.

The system was able to gather relevant data from the users, reason through the IF- THEN type of rules, and then suggest laptop options that fit the users' preferences.

It is hoped that this project will provide an introductory-level understanding of how facts, inference rules, and decision-making can work together if they are integrated in a computer-based system that is an expert system.